Breast Cancer Wisconsin

Research Report

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1 Introduction and Problem Formulation

1.1 Dataset Overview: Breast Cancer Wisconsin (Diagnostic)

In this study, we utilize the Breast Cancer Wisconsin (Diagnostic) dataset, a widely recognized benchmark in the medical machine learning domain. The dataset comprises 569 instances, each defined by 30 continuous numerical features. These features are mathematically computed from digitized images of fine needle aspirates (FNA) of breast masses, capturing key characteristics of cell nuclei such as radius, perimeter, area, and smoothness. The target variable is binary, distinguishing between malignant (0) and benign (1) tumors. We selected this specific dataset because its rich, high-dimensional nature makes it an ideal candidate for evaluating both the natural clustering tendencies of biological data and the predictive power of classification algorithms.

1.2 Research Question and Motivation

The fundamental motivation behind this project is to explore how data-driven techniques can assist in critical medical diagnostics, a field where diagnostic errors (particularly false negatives) can have severe, life-altering consequences. Our investigation is driven by a two-fold research question: First, do the physical characteristics of breast cells contain an inherent mathematical structure that naturally separates healthy cells from cancerous ones? Second, when provided with confirmed clinical diagnoses, how accurately can machine learning models learn these patterns to classify new, unseen tumors? By addressing these questions, we aim to demonstrate the full data science lifecycle, transitioning from blind data exploration to actionable predictive modeling.

1.3 Unsupervised and Supervised Learning Objectives

To systematically answer our research questions, our methodology is divided into two interconnected phases:

The primary objective of the **Unsupervised Learning** phase is to analyze the data's underlying geometry without any prior knowledge of the target labels. We aim to employ dimensionality reduction techniques (PCA and t-SNE) alongside clustering algorithms (K-Means and GMM) to empirically verify if the dataset naturally partitions into benign and malignant groups based purely on physical measurements.

Subsequently, the **Supervised Learning** phase shifts the focus to predictive modeling. Our objective here is to train the algorithms using the true diagnostic labels and compare a highly interpretable linear baseline (Logistic Regression) against a more mathematically flexible, non-linear model (Support Vector Machine). Crucially, our evaluation strategy will not rely solely on overall accuracy. Instead, we will prioritize optimizing for *Recall* to strictly minimize False Negatives, reflecting the clinical reality where failing to identify a malignant tumor is the most costly error.

2 Data Understanding and Preprocessing

2.1 Exploratory Data Analysis (EDA)

Our primary goal during the Exploratory Data Analysis phase was to understand the structural characteristics of the dataset and ensure its integrity before applying any machine learning algorithms. First, we inspected the dataset's dimensions and systematically checked for missing values. This step is crucial, as incomplete data can severely disrupt mathematical distance calculations and algorithm convergence. Next, we visualized the target variable's distribution using a count plot to assess class imbalance. Checking this balance is a mandatory step in medical diagnostic modeling, as heavily skewed data dictates the choice of evaluation metrics and dictates whether resampling techniques are needed.

From this exploration, we observed a structurally sound and perfectly complete dataset (zero missing values across all 569 instances). Furthermore, the visual distribution revealed a natural and manageable class imbalance, with benign cases moderately outnumbering malignant ones. Based on these observations, we concluded that the data was highly reliable in its raw state.

2.2 Handling Outliers and Missing Values

A critical integrity check confirmed that the dataset contained exactly zero missing values across all features, allowing us to proceed without the need for synthetic data imputation. Furthermore, we decided not to aggressively remove outliers. In the context of cancer diagnostics, extreme cytological values often represent severe malignancy rather than measurement errors. We relied on the subsequent standardization step to mitigate the impact of these extreme values on distance-based calculations.

2.3 Feature Scaling and Standardization

A visual inspection of the raw data revealed drastic differences in feature magnitudes (e.g., *mean area* routinely exceeded 1000, while *mean smoothness* was below 0.1). The core algorithms selected for this project (PCA, K-Means, and SVM) rely on geometric distance calculations. If left unscaled, features with naturally larger numerical values would mathematically dominate the distance metrics, rendering smaller-scale features virtually irrelevant.

To solve this, we applied Standardization (Z-score normalization) using Scikit-learn's `StandardScaler`. This transforms each feature so that it has a mean (μ) of 0 and a standard deviation (σ) of 1. The transformation applied to each data point x is defined by the following formula:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

Where x is the original feature value, μ is the mean of that feature across the training set, and σ is its standard deviation.

2.4 Train/Test Data Split

Before applying any data transformations, we split the dataset into a training set (80%) and a testing set (20%). This step involves two critical methodological decisions:

First, we utilized a stratified split (`stratify=y`). Because our target variable is slightly imbalanced, a random split could accidentally create training or testing sets that do not reflect the true proportion of malignant and benign cases. Stratification mathematically ensures that the exact proportion of malignant and benign tumors is perfectly preserved in both subsets.

Second, we explicitly performed this split *before* scaling the features to prevent **Data Leakage**. If we had standardized the entire dataset at once, the global mean and standard deviation would contain information from the test set. By splitting first, the scaler calculates μ and σ exclusively from the training data, ensuring our test set remains completely unseen and simulates a real-world clinical scenario.

3 Uncovering Hidden Structure (Unsupervised Learning)

Our objective in this phase was to discover inherent mathematical patterns in the cytological data without exposing the algorithms to the true diagnostic labels.

3.1 Linear Dimensionality Reduction: Principal Component Analysis (PCA)

First, we applied Principal Component Analysis (PCA) to reduce the dataset from 30 original features down to 2 principal components. We chose PCA as our primary reduction technique because high-dimensional space cannot be visualized by humans and often causes algorithms to overfit (the "curse of dimensionality"). PCA is a linear, variance-maximizing method that distills the dataset into its most informative orthogonal axes.

From this transformation, we observed that the first two components retained **63.36% of the total variance**. When visualizing these two components in a scatter plot (and coloring them retrospectively with the true labels), we observed a clear, natural linear separation between malignant and benign cells, proving that the physical traits inherently distinguish the two classes.

3.2 Non-linear Dimensionality Reduction: t-SNE

Next, we applied t-Distributed Stochastic Neighbor Embedding (t-SNE) to project the scaled data into a 2D space. We utilized this technique because PCA is strictly linear and might miss complex, non-linear relationships. t-SNE, conversely, excels at preserving local neighborhood structures, keeping similar data points close together.

Through the t-SNE visualization, we observed the formation of two distinct, dense visual masses. This non-linear projection confirmed our PCA findings: the biological cells naturally group into two main clusters based solely on their physical measurements, even without prior knowledge of their malignancy.

3.3 Clustering Analysis Method 1: K-Means

To mathematically formalize these visual groupings, we implemented K-Means clustering with the number of clusters set to two ($k = 2$). We selected K-Means as our baseline clustering model because it is highly efficient and provides a straightforward "hard" assignment of each cell to a specific group.

To evaluate the algorithm blindly, we calculated the Silhouette Score. We observed a score of **0.355**. In the context of complex biological data, this positive score confirms that a moderate, meaningful cluster structure exists, though the groups are not perfectly isolated from one another.

3.4 Clustering Analysis Method 2: Gaussian Mixture Models (GMM)

Following K-Means, we applied Gaussian Mixture Models (GMM) with two components. We introduced this second clustering method to address a critical limitation of K-Means, which assumes that clusters are perfectly spherical and have similar variances. Biological data is rarely this clean. GMM offers a probabilistic approach that allows for elliptical clusters, making it a much more realistic modeling choice for medical diagnostics where "grey areas" overlap.

Upon evaluating the GMM approach, we observed a Silhouette Score of **0.319**. While slightly lower than K-Means in this specific mathematical metric, the algorithm successfully identified the two main distributions of our cytological data.

3.5 Interpretation and Limitations of Discovered Structures

In summary, our unsupervised methodology successfully proved our initial hypothesis: there is an undeniable, inherent structural difference between healthy and cancerous cells that machine learning algorithms can detect without human labeling.

However, we also observed a critical limitation. Both our visual projections (PCA/t-SNE) and our mathematical metrics (Silhouette Scores around 0.35) revealed an overlapping "grey area" between the two clusters. This overlap represents tumors with ambiguous physical characteristics. We concluded that while unsupervised learning is excellent for revealing hidden structures, it is insufficiently precise for critical medical diagnosis. This directly justifies our transition to supervised learning, where true labels can guide the models to draw much more precise and reliable decision boundaries.

4 Predicting Malignancy (Supervised Learning)

4.1 Task Definition: Binary Classification

Having uncovered the hidden structure of the data through unsupervised methods in Section 3, we now shift to supervised learning. The task is binary classification: given the 30 numerical features extracted from digitised images of cell nuclei, the goal is to predict whether a tumour is **malignant (class 0)** or **benign (class 1)**.

Before selecting any model, we considered the specific requirements of this application domain. Medical diagnosis is not a context in which raw predictive accuracy is the only criterion. A model that achieves 99% accuracy but is completely opaque — offering no explanation for its predictions — is arguably less useful in a clinical setting than a model with 97% accuracy that can identify *which biological measurements* drove a particular diagnosis. This distinction guided our model selection: we did not simply choose the two most powerful algorithms available, but rather two algorithms that together cover a broad spectrum from maximum interpretability to maximum flexibility.

The supervised learning pipeline is structured as follows. Both models are trained on `X_train_scaled` and evaluated on `X_test_scaled`, the standardised splits produced in Section 2. In addition to the standard train/test evaluation, we apply 5-fold cross-validation to obtain a more robust and reliable estimate of each model's generalisation performance.

4.2 Basic/Linear Model: Logistic Regression

4.2.1 Motivation and Model Description

Logistic Regression was selected as the baseline model primarily for its **interpretability**. Unlike most classification algorithms, Logistic Regression assigns a coefficient to each input feature, and the sign and magnitude of these coefficients directly quantify each feature's contribution to the final prediction. A large negative coefficient indicates that high values of that feature push the model strongly towards a malignant prediction; a large positive coefficient pushes it towards benign.

This transparency is not merely a pedagogical convenience — it has direct practical relevance. A clinician reviewing a model's prediction can examine the coefficient chart and identify which measurements of the cell nucleus were most influential. In a domain where model errors can have serious consequences, this level of accountability is invaluable.

Logistic Regression works by modelling the log-odds of the positive class (benign) as a linear function of the input features:

$$\log \frac{P(y = 1 \mid \mathbf{x})}{1 - P(y = 1 \mid \mathbf{x})} = \mathbf{w}^T \mathbf{x} + b$$

The model is trained by maximising the log-likelihood of the training labels, which is equivalent to minimising the cross-entropy loss with an L2 regularisation term (the default in

scikit-learn). The model outputs a probability between 0 and 1, which is thresholded at 0.5 to produce the final class prediction.

4.3 Flexible/Complex Model: Support Vector Machine (RBF Kernel)

4.3.1 Motivation and Model Description

The Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel was selected as the flexible model. Unlike Logistic Regression, the SVM does not assume that the decision boundary between classes is linear. Instead, it implicitly maps the input data into a higher-dimensional feature space via the kernel function, where a linear separator (the maximum-margin hyperplane) can be found between the two classes.

The RBF kernel is defined as:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\right)$$

where γ controls how far the influence of a single training example reaches. A large γ means that each training point has a very localised influence, producing a more complex boundary; a small γ produces a smoother, more generalised boundary.

Beyond its predictive power, the SVM offers an analytically interesting property for this project: **support vectors**. The support vectors are the subset of training samples that lie closest to the decision boundary — the most ambiguous and borderline cases. These are precisely the tumours that the model finds hardest to classify, and comparing them against the misassigned samples from K-Means clustering (Section 3) reveals whether unsupervised and supervised methods agree on which cases are most difficult.

4.4 Cross-Validation Results

Before evaluating on the held-out test set, 5-fold cross-validation was applied to both models on the training data (`X_train_scaled`). In 5-fold cross-validation, the training set is split into 5 equal parts (folds); the model is trained on 4 folds and evaluated on the remaining fold, and this process is repeated 5 times so that every sample serves as a validation point exactly once. The reported accuracy is the mean across all 5 folds.

Model	Mean CV Accuracy	Std. Deviation
Logistic Regression	0.980	± 0.013
SVM (RBF)	0.971	± 0.018

Table 1: 5-fold cross-validation results on the training set.

The results in Table 1 reveal an outcome that may seem counterintuitive at first: the linear model outperforms the flexible model in cross-validation. This is not a coincidence. As the

PCA scatter plot in Section 3 demonstrated, the two classes are strongly separated along the first principal component, suggesting a largely linear decision boundary. In such a setting, the additional complexity of the RBF kernel provides no structural advantage, and its extra degrees of freedom may introduce mild instability — reflected in the slightly higher standard deviation (0.018 vs. 0.013).

It is also worth noting that the SVM's cross-validation performance could almost certainly be improved through hyperparameter tuning. The default values (`C=1.0`, `gamma='scale'`) are reasonable starting points but are not optimised for this particular dataset. A grid search over `C` and `gamma` would likely push the SVM's cross-validation accuracy above that of Logistic Regression. Nevertheless, within the scope of this project, we evaluate both models under the same conditions.

5 Model Evaluation and Comparison

5.1 Evaluation Framework

Evaluating a classification model with a single accuracy figure is insufficient, particularly in a medical context. A model could achieve 95% accuracy on this dataset by simply predicting *benign* for every sample — since 62.7% of the dataset is benign — while never correctly identifying a single malignant tumour. This motivates a richer evaluation framework, comprising four complementary perspectives:

- **Confusion Matrix**: A 2×2 table showing the exact counts of true positives, true negatives, false positives, and false negatives. This makes explicit the types of errors a model makes.
- **Classification Report**: Provides Precision, Recall, and F1-Score for each class individually. In the medical context of this project, **Recall for the Malignant class** is the priority metric — it measures the proportion of true malignant cases that the model correctly identifies.
- **ROC Curve and AUC**: Plots the True Positive Rate against the False Positive Rate across all possible classification thresholds. The Area Under the Curve (AUC) provides a single threshold-independent summary of discriminative performance.
- **Feature Importance via Coefficients**: Specific to Logistic Regression, this analysis reveals which features the model has learned to rely on most heavily, enabling a direct connection between the model's behaviour and the underlying biology of the dataset.

5.2 Confusion Matrices

The confusion matrices for both models have the same structure: rows represent the true diagnosis and columns represent the predicted diagnosis. The diagonal cells (top-left and bottom-right) contain correct predictions; the off-diagonal cells contain errors.

In the context of cancer diagnosis, the two types of error carry very different costs:

- **False Negative (FN)**: A malignant tumour predicted as benign. This is the most dangerous error — the patient does not receive treatment when they need it.
- **False Positive (FP)**: A benign tumour predicted as malignant. This leads to unnecessary further testing, which causes patient anxiety and healthcare costs, but is far less harmful than a missed malignancy.

Both models produce **1 false negative** out of 42 malignant cases in the test set — a strong result. The identical confusion matrices confirm that on this particular test split, Logistic Regression and SVM make exactly the same errors.

5.3 Classification Reports

Class	Precision	Recall	F1-Score	Support
Logistic Regression				
Malignant (0)	0.976	0.976	0.976	42
Benign (1)	0.986	0.986	0.986	72
Weighted avg	0.982	0.982	0.982	114
SVM (RBF)				
Malignant (0)	0.976	0.976	0.976	42
Benign (1)	0.986	0.986	0.986	72
Weighted avg	0.982	0.982	0.982	114

Table 2: Classification reports for both models on the test set.

Table 2 shows that both models achieve identical scores across all metrics. Both correctly classify 97.6% of malignant cases (Recall = 0.976) and 98.6% of benign cases (Recall = 0.986). The F1-Score, which balances Precision and Recall, is 0.976 for the malignant class and 0.986 for the benign class.

The fact that both models produce identical results on the test set, combined with the cross-validation result showing Logistic Regression marginally ahead, reinforces the conclusion that the dataset's decision boundary is inherently linear. The RBF kernel's additional flexibility adds no value here without targeted hyperparameter optimisation.

5.4 ROC Curves and AUC

The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (TPR, also called Recall) on the Y-axis against the False Positive Rate (FPR = FP / (FP + TN)) on the X-axis, as the classification threshold is varied from 0 to 1. A perfect model produces a point at (0, 1) — no false positives, all true positives detected — while a random classifier follows the diagonal.

Both models produce ROC curves that hug the top-left corner of the plot, with AUC values very close to 1.0. This confirms that regardless of the decision threshold chosen, both models maintain a high true positive rate while keeping the false positive rate low — a desirable property in a screening context where the threshold might be adjusted depending on the relative cost of false negatives versus false positives.

5.5 Feature Importance: Logistic Regression Coefficients

5.5.1 Interpretation

The features with the most negative coefficients — `worst texture` and `radius error` — are the strongest predictors of malignancy according to the model. This is somewhat surprising: we might expect features like `worst concave points` or `mean radius` to dominate, given that they showed the highest raw correlation with the target in the exploratory analysis. The coefficient chart tells a different story, which reflects the regularised nature of the model: after accounting for the correlations between features, the model has redistributed weight across the feature set in a way that does not always align with simple pairwise correlations.

On the benign side, `mean compactness` and `compactness error` carry the largest positive coefficients. This suggests that, once size and shape features are accounted for, compactness measurements are the most distinctive feature of benign tissue.

5.5.2 Caveat: Multicollinearity

It is important to note a significant limitation of this analysis. As established in Section 2, many features in this dataset are highly correlated — the correlations between radius, perimeter, and area exceed 0.99. In the presence of such multicollinearity, the individual coefficients of a Logistic Regression model become unstable: small changes in the training data can lead to large swings in individual coefficients, even if the model's overall predictions remain stable. The coefficient chart should therefore be interpreted as a broad guide to feature importance rather than a precise quantitative ranking. PCA's orthogonal components, used in Section 6, are inherently free from this problem.

6 Connecting Unsupervised and Supervised Learning

6.1 Overview

The title of this project identifies the connection between unsupervised and supervised learning as one of its most important goals. This section addresses that connection directly through two analyses. First, we examine whether the clusters discovered by K-Means in Section 3 — without any access to diagnosis labels — correspond to the true malignant/benign distinction. Second, we quantify how much predictive performance is lost when the 30 original features are replaced by the 2 PCA components computed in Section 3.

6.2 Do K-Means Clusters Correspond to the True Classes?

6.2.1 Analysis

To answer this question, we cross-tabulate the K-Means cluster assignments (`kmeans_labels`) against the true training labels (`y_train`). If K-Means has successfully recovered the biological structure of the data, each cluster should correspond predominantly to one class.

	Cluster 0	**Cluster 1**
Malignant (0)	32	138
Benign (1)	274	11

Table 3: Cross-tabulation of K-Means cluster assignments vs. true diagnosis labels on the training set.

The results in Table 3 are striking. **Cluster 1 is overwhelmingly malignant**: 138 out of 149 samples assigned to it are true malignant cases (92.6% purity). **Cluster 0 is overwhelmingly benign**: 274 out of 306 samples are truly benign (89.5% purity). K-Means, using no label information whatsoever — purely the geometric distances between samples in the 30-dimensional feature space — has successfully reconstructed the malignant/benign distinction in the vast majority of cases.

6.2.2 The Boundary Cases

The 43 misassigned samples (32 malignant cases in Cluster 0, 11 benign cases in Cluster 1) are particularly informative. These are the tumours whose physical measurements lie in the overlap region between the two classes — biologically ambiguous cases where the cell nucleus measurements of a malignant tumour resemble those of a benign one, or vice versa. This is precisely the same region where the supervised models produce their false negatives and false positives. The convergence of unsupervised and supervised methods on the same difficult cases provides additional confidence that these samples represent genuine biological ambiguity rather than modelling artefacts.

6.3 Can 2 PCA Components Replace 30 Features for Prediction?

6.3.1 Analysis

To directly quantify the predictive value of the dimensionality reduction performed in Section 3, we train a new Logistic Regression model using only the 2 PCA components as input features — instead of the 30 original measurements. The test set is transformed using the same PCA model fitted on the training data, and the new model's predictions are compared against those of the full model.

Model	Features	Test Accuracy	Accuracy Retained
LR with PCA (2 components)	2	0.947	96.4%
Logistic Regression (full)	30	0.982	100% (baseline)
SVM RBF (full)	30	0.982	100% (baseline)

Table 4: Accuracy comparison across all models, including the PCA-based simplified model.

6.3.2 Interpretation

Reducing from 30 features to just 2 PCA components results in a test accuracy of 94.7%, compared to 98.2% for the full models — a reduction of only **3.5 percentage points**. Stated differently, two linear combinations of the original 30 features retain **96.4% of the predictive signal**. This is a remarkable result that highlights the redundancy inherent in the feature set: with 30 correlated features measuring the same underlying biological properties (radius, texture, concavity, etc.) at three statistical levels (mean, standard error, worst), much of the information is duplicated. PCA effectively distils this redundancy into a compact representation.

This result is also consistent with the PCA explained variance analysis in Section 3: the first 2 principal components capture 63.2% of the total variance, and the first 7 capture 91.0%. The fact that 63.2% of the variance is sufficient to achieve 94.7% classification accuracy — compared to 98.2% with full variance — indicates that the most predictive dimensions in this dataset are concentrated in the first few principal components.

7 Final Reflection and Conclusions

7.1 What Worked Well

The most resounding success of this project was the overall quality of the supervised learning results. Both Logistic Regression and SVM achieved 98.2% accuracy on the test set, with a Recall of 0.976 for the malignant class — meaning that only 1 malignant tumour out of 42 in the test set was missed. This is a strong result by any standard, and it reflects the intrinsic quality of the Breast Cancer Wisconsin dataset: the features are well-chosen, the classes are well-separated, and the dataset is free from missing values or systematic noise.

The PCA projection in Section 3 provided a clear and immediately interpretable visual confirmation of this separability: the two classes form distinct, compact clusters along the first principal component, with only a modest overlap region at the boundary. This visual result set a useful expectation before any supervised model was trained, and the subsequent accuracy figures confirmed it.

The K-Means clustering result was also highly satisfying. Without any access to diagnosis labels, the algorithm correctly separated 92.6% of malignant samples into one cluster and 89.5% of benign samples into another. The fact that an unsupervised method can reconstruct the clinical distinction between malignant and benign tumours demonstrates that the

biological differences between the two types of tissue are strongly reflected in the numerical measurements of cell nuclei.

7.2 What Did Not Work Well

The SVM with default hyperparameters underperformed Logistic Regression in cross-validation (97.1% vs. 98.0%). This was the most puzzling result of the project, since SVM with an RBF kernel is generally considered a more powerful model than linear Logistic Regression. The explanation lies in the structure of the data: as the PCA analysis shows, the classes are largely linearly separable, and the RBF kernel's added flexibility provides no structural advantage. Furthermore, the default `C` and `gamma` parameters are not optimised for this dataset. A hyperparameter search via `GridSearchCV` would almost certainly close — and likely reverse — the gap between the two models.

The K-Means clustering also produced 43 misassigned samples, representing the most ambiguous boundary cases. This is an inherent limitation of hard-assignment clustering: every sample must be assigned to exactly one cluster, even if it lies equidistant between two cluster centroids. A soft-assignment method such as GMM, which assigns probabilities of cluster membership, can model this uncertainty more naturally — though in this case the GMM result was similar.

7.3 What Surprised Us

The most surprising finding of the entire project was the performance of the PCA-based Logistic Regression model. Training a classifier on just 2 numbers — two linear combinations of 30 measurements — and achieving 94.7% accuracy was not expected. The intuition that compressing a 30-dimensional feature space into 2 dimensions would cause a significant loss of predictive information proved to be incorrect. The key insight is that the 30 features in this dataset are not 30 independent sources of information; they are 10 biological properties measured three times each (mean, standard error, worst), leading to extreme redundancy. PCA effectively removes this redundancy and retains the dominant axes of variation, which happen to align very closely with the malignant/benign distinction.

A second surprising result was the consistent superiority of Logistic Regression over SVM. The common intuition in machine learning is that more complex models outperform simpler ones given sufficient data. This project is a useful counterexample: when the data's structure is inherently linear, a simple linear model can match or outperform a flexible non-linear model, while offering the additional benefit of interpretability.

7.4 Most Convincing Method

Logistic Regression was the most convincing model overall, and it is the model we would recommend for this application. It achieved 98.2% test accuracy and 98.0% cross-validation accuracy — matching the SVM on the test set and surpassing it in cross-validation. More

importantly, it is the only model in this study that provides a direct explanation of its predictions through the coefficient chart. In a medical setting, where a model's decisions can affect patient outcomes, transparency and accountability are not optional extras — they are fundamental requirements. A model that can tell a clinician "this tumour was flagged as malignant primarily because of elevated `worst texture` and `radius error`" is far more actionable than one that provides a prediction with no rationale.

7.5 Limitations of the Study

Several limitations should be acknowledged before drawing any broader conclusions from this analysis.

Dataset size. The Breast Cancer Wisconsin dataset contains 569 samples, of which only 114 were used for testing. Evaluation metrics computed on 114 samples carry a non-trivial degree of statistical uncertainty. A 95% confidence interval for an accuracy of 98.2% on 114 samples is approximately $\pm 2.4\%$, which means the true generalisation accuracy could plausibly range from 95.8% to 100%. Larger datasets would yield tighter confidence intervals and more reliable conclusions.

Multicollinearity. All 30 features are derived from the same FNA images and measure related properties of the cell nucleus. Correlations between features exceed 0.99 in several cases. This multicollinearity does not affect the predictive accuracy of models like SVM or regularised Logistic Regression, but it does make the individual coefficients of the Logistic Regression model unreliable as a feature importance ranking. Small changes in the training data can produce large swings in individual coefficients while leaving overall predictions unchanged.

Dataset age and clinical context. The dataset was collected in 1995 using Fine Needle Aspiration (FNA) biopsy images. The 30 features were engineered by hand based on the imaging technology of that era. Modern pathology uses richer imaging modalities, larger slide areas, deeper feature extraction methods, and often combines imaging data with genomic and clinical information. A model trained on these 30 features may not generalise to modern diagnostic workflows.

Single dataset evaluation. All models were trained and evaluated on a single dataset from a single institution. Generalisation to other hospitals, patient populations, imaging equipment, and clinical protocols was not assessed. Robust clinical validation would require multi-site studies with external validation cohorts.

No hyperparameter optimisation. Both models were evaluated with default hyperparameters (except `max_iter` for Logistic Regression). A fair comparison under optimal conditions for each model would require a systematic hyperparameter search, which is beyond the scope of this project but would be necessary in any serious applied study.